

Report 2

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1 Attempt to Denoise

The audio data that is recorded is filled with external noise, like engine growl, people speaking and such. To combat this a simple de-noising scheme was implemented, pre-emphasizing the noise followed by FFT unwanted frequency reduction. This reduces the noise to signal ratio. But overall prediction suffers because **YAMNet** is trained on extremely noisy data. Will try with other types of pre-trained network.

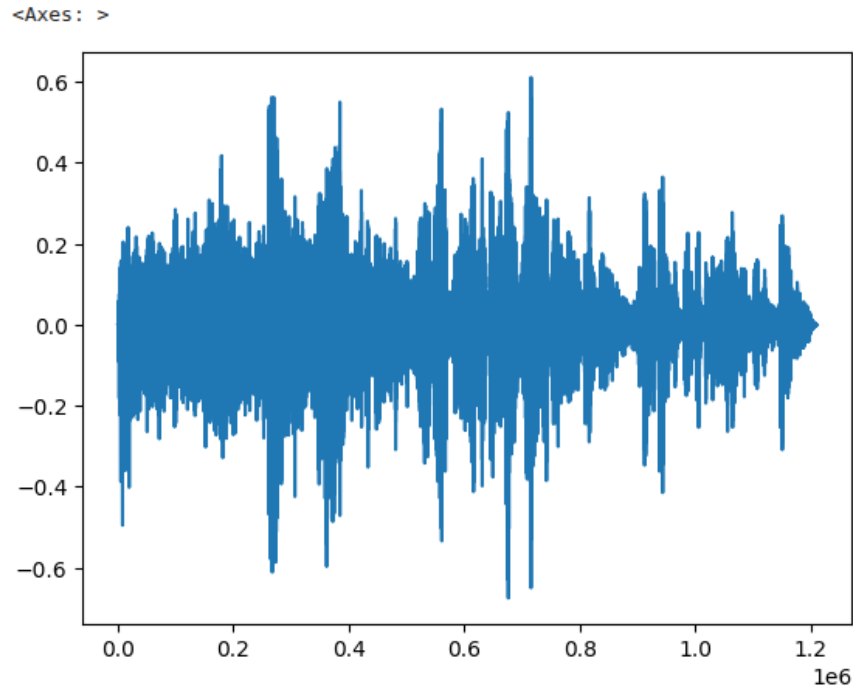


Figure 1: Original

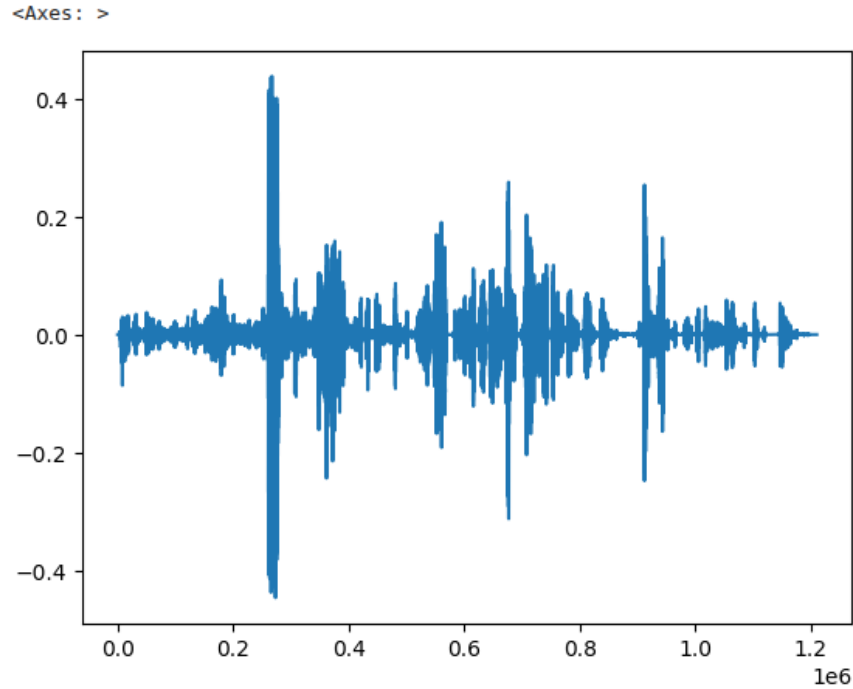


Figure 2: Denoised

2 Filtering and PCA with $n_components > 2$

To improve the prediction capabilities of the system, I impletemend filtering as discussed, removing the *Transformer* from the loop to avoid haluciantion. A Mode based measure was utilized to predict if the value is actually a horn.

To improve the chances of prediction we include other unrelated values aswell, *Siren*, *Alarm*, *Buzzer*. To increase the broad classification range.

Principle component analysis works extrmely well for n.components with a value of 5.

3 K-Means and HDBSCAN

K-Means Clustering performed better than the original Birch alorithm. HDBSCAN was also able to identify values which are not classifiable. Marked out with the value of -1.

The graphs are based on PCA-2 so that we can display the graphs.

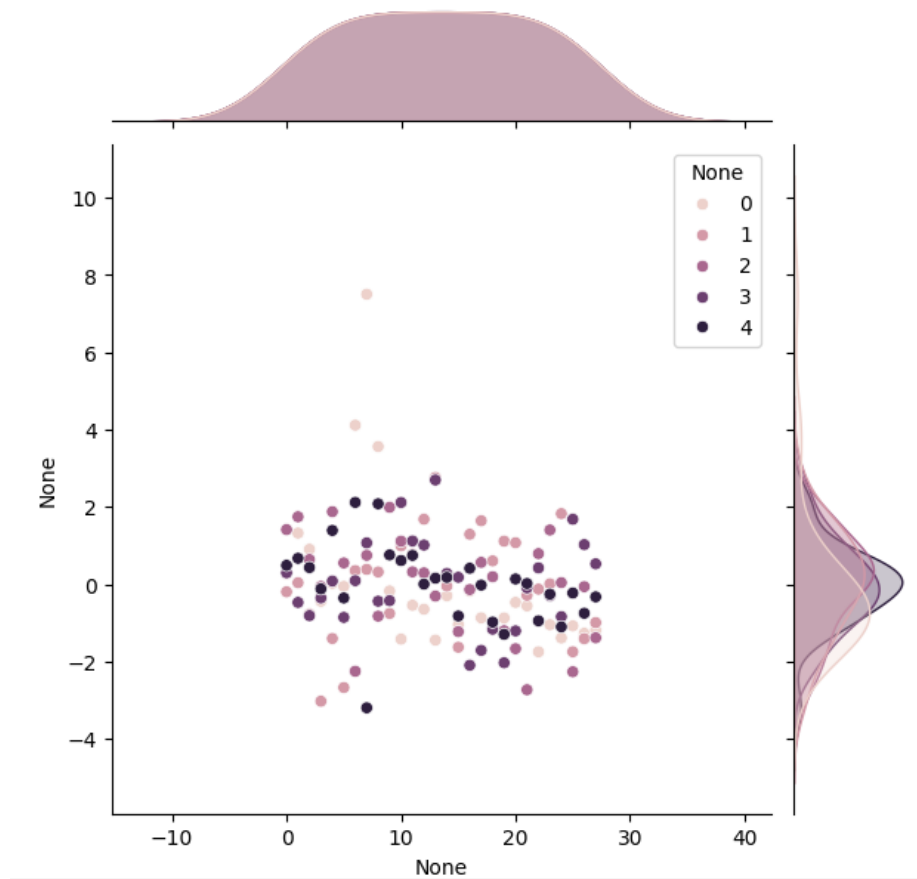


Figure 3: PCA with $n_components = 5$

4 Labeling of data

A bunch of data point have been labeled, I downloaded data from *freesound.org*, *BBC Sound* and Other manufacturer sites. I will label further.

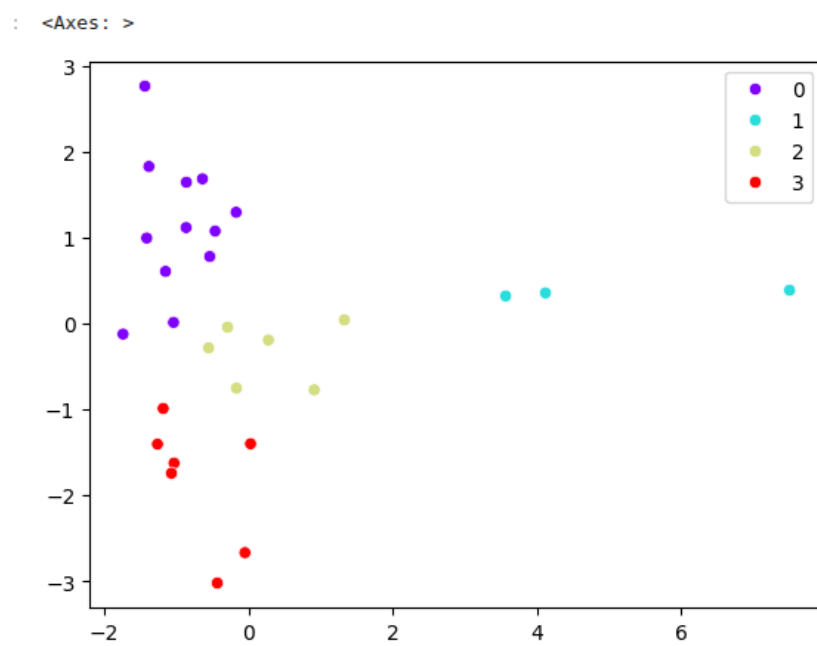


Figure 4: K-means based on PCA-2

